Mathematical and Computational Methods

Methods

Part IV: Integration (Units 9–10)

Most problems on the exercise sheets are instances of a few standard types, and each type has a *method*: a fixed sequence of steps, taken from the notes, that solves every problem of that type. This guide collects them, unit by unit. Each method states the problem it solves, lists its steps and works one example that is not on any sheet. It ends with the mistakes to avoid, where the notes teach it, and a link to a Review set of problems of that kind. The units fall into four parts: foundations and complex numbers, vectors and matrices, differentiation, and integration.

The guide does not say which sheet problem needs which method. Recognising the type is part of what the sheets train, and in an exam nothing tells you. Where a unit's problems look alike but need different methods, its chapter opens with a table for choosing.

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Part IV Integration

Units 9–10: the definite integral, the fundamental theorem and substitution; then the techniques for the integrands substitution cannot reach, and the integrals that model an area, a volume, work, a mass or a probability. Try a substitution first. Unit 10's table says what to try next.

Unit 9 Integration I: the definite integral and substitution

Six methods. The table of antiderivatives comes straight after Riemann sums, because FTC 2 needs an antiderivative; the rest follow the order of the notes. The properties of the definite integral (splitting the interval, reversing the limits, linearity) are single moves inside them. The last method is for the integrals the others cannot reach: a continuous function such as e^{-x^2} has an antiderivative, but not an elementary one, and its definite integral is bracketed by Riemann sums.

9.1 Riemann sums and their limit

Problem. Estimate $\int_a^b f(x)\,dx$ by a Riemann sum with n strips, and say whether it is an over- or an under-estimate; for a polynomial f of degree at most two, find the right sum for general n in closed form and take its limit.

Method

1. Cut $[a,b]$ into n strips of width $\Delta x = \dfrac{b-a}{n}$. The division points are $x_k = a + k\,\Delta x$: right endpoints for $k = 1,\dots,n$, left endpoints for $k = 0,\dots,n-1$.
2. Add the rectangles, $\sum f(x_k)\,\Delta x$, listing the heights $f(x_k)$.
3. If f increases on the whole of $[a,b]$, the right sum over-estimates and the left sum under-estimates; if it decreases, it is the other way round. Either way the exact value is trapped between the two. If f turns inside $[a,b]$, the two sums need not trap it.
4. For general n, take the right sum, $S_n = \sum_{k=1}^{n} f(a+k\,\Delta x)\,\Delta x$. Write it as a sum in k, expand, and pull the powers of n out of the sums.
5. Quote $\sum_{k=1}^{n} k = \frac{1}{2}n(n+1)$ and $\sum_{k=1}^{n} k^2 = \frac{1}{6}n(n+1)(2n+1)$, and simplify to a closed form in n.
6. Let $n \to \infty$, and compare the limit with the value from FTC 2 (Method 9.3).

Example. Let $f(x) = 6 - x - x^2$ on $[0,2]$. (a) Estimate $\int_0^2 f(x)\,dx$ with $n = 4$ strips, by the right and the left sums. Which is the over-estimate? (b) Find the right sum S_n with n strips in closed form, and its limit.

(a) $\Delta x = \frac{2-0}{4} = 0.5$. The heights at $x = 0,\ 0.5,\ 1,\ 1.5,\ 2$ are $6,\ 5.25,\ 4,\ 2.25,\ 0$, so

$$\text{right} = 0.5\,(5.25 + 4 + 2.25 + 0) = 5.75,$$

$$\text{left} = 0.5\,(6 + 5.25 + 4 + 2.25) = 8.75.$$

f is decreasing on $[0,2]$, since x and x^2 both increase there. So the left end is the tallest point

of each strip: the left sum is the over-estimate, and

$$\boxed{5.75 < \int_0^2 f(x)\,\mathrm{d}x < 8.75}$$

(b) Now $\Delta x = \frac{2}{n}$ and $x_k = \frac{2k}{n}$, so $f(x_k) = 6 - \frac{2k}{n} - \frac{4k^2}{n^2}$ and

$$\begin{aligned} S_n &= \sum_{k=1}^{n}\left(6 - \frac{2k}{n} - \frac{4k^2}{n^2}\right)\frac{2}{n} \\ &= 12 - \frac{4}{n^2}\sum_{k=1}^{n} k - \frac{8}{n^3}\sum_{k=1}^{n} k^2 \\ &= 12 - \frac{2(n+1)}{n} - \frac{4(n+1)(2n+1)}{3n^2}. \end{aligned}$$

Since $\frac{2(n+1)}{n} = 2 + \frac{2}{n}$ and $\frac{4(n+1)(2n+1)}{3n^2} = \frac{8n^2+12n+4}{3n^2} = \frac{8}{3} + \frac{4}{n} + \frac{4}{3n^2}$,

$$\boxed{S_n = \frac{22}{3} - \frac{6}{n} - \frac{4}{3n^2}}$$

As $n \to \infty$ the last two terms vanish, and

$$\int_0^2 f(x)\,\mathrm{d}x = \lim_{n\to\infty} S_n = \boxed{\tfrac{22}{3}}.$$

Check. $n = 4$ gives $S_4 = \frac{22}{3} - \frac{3}{2} - \frac{1}{12} = \frac{23}{4} = 5.75$, the right sum of (a). FTC 2 gives

$$\left[6x - \tfrac{1}{2}x^2 - \tfrac{1}{3}x^3\right]_0^2 = 12 - 2 - \tfrac{8}{3} = \tfrac{22}{3} \approx 7.33,$$

inside the bracket. And $S_n < \frac{22}{3}$ for every n, as an under-estimate must be.

Pitfalls

- Calling the right sum an over-estimate out of habit. That holds only on an increasing function; here f decreases and the right sum is the under-estimate.
- Summing over $k = 0, \ldots, n$: that is $n + 1$ rectangles, one of them outside $[a, b]$.
- Writing $x_k = \frac{k}{n}$ on an interval that is not $[0, 1]$. On $[0, 2]$ the points are $x_k = \frac{2k}{n}$, and the width is $\frac{2}{n}$.

In the notes. Unit 9: The area under a curve, and Unit 9: Computing an integral from the definition.

Practise it. the Review set on this method

9.2 Antiderivatives from the table: rewrite, then read off a

Problem. Find an antiderivative of a sum of powers, roots, reciprocals and the table's entries e^{ax}, $\sin ax$, $\cos ax$, $\sec^2 ax$, $\frac{1}{a^2+x^2}$ and $\frac{1}{\sqrt{a^2-x^2}}$, as an indefinite integral or as the F that FTC 2 needs.

Method

1. Rewrite the integrand as a sum of powers and table entries. Write roots and reciprocals as powers ($\sqrt{x} = x^{1/2}$, $\frac{1}{x^2} = x^{-2}$), split a quotient by a one-term denominator term by term (a sum in the denominator does not split), and take constant multiples out.
2. Integrate each power by the power rule, $x^n \to \dfrac{x^{n+1}}{n+1}$ for $n \neq -1$. The exception, $\frac{1}{x}$, gives $\ln|x|$.
3. Match every other term to a table row and read off $a > 0$. In e^{ax}, $\sin ax$, $\cos ax$ and $\sec^2 ax$, a is the coefficient of x, and the answer carries $\frac{1}{a}$ (with a minus sign for sin).
4. In $\frac{1}{a^2+x^2}$ and $\frac{1}{\sqrt{a^2-x^2}}$, the constant is a^2, so a is its square root. The arctan row carries $\frac{1}{a}$; the arcsin row has none.
5. Add a single $+C$. For a definite integral, leave out the $+C$ and go on to Method 9.3.
6. Check by differentiating the answer term by term.

Example. Find (a) $\displaystyle\int \frac{x^3 - 2\sqrt{x} + 5}{x}\,dx$ and (b) $\displaystyle\int \left(3\sin\frac{x}{2} + \frac{6}{3+x^2}\right)dx$.

(a) Divide term by term: the integrand is $x^2 - 2x^{-1/2} + \frac{5}{x}$. The powers give $\frac{1}{3}x^3$ and $-2 \cdot \frac{x^{1/2}}{1/2} = -4x^{1/2}$, and $\frac{5}{x}$ gives $5\ln|x|$:

$$\boxed{\tfrac{1}{3}x^3 - 4\sqrt{x} + 5\ln|x| + C}$$

(b) In $\sin\frac{x}{2}$, $a = \frac{1}{2}$, so $\frac{1}{a} = 2$ and $\int 3\sin\frac{x}{2}\,dx = -6\cos\frac{x}{2}$. In $\frac{6}{3+x^2}$, $a^2 = 3$, so $a = \sqrt{3}$, and the term gives $6 \cdot \frac{1}{\sqrt{3}} \arctan\frac{x}{\sqrt{3}} = 2\sqrt{3}\arctan\frac{x}{\sqrt{3}}$. Hence

$$\boxed{-6\cos\tfrac{x}{2} + 2\sqrt{3}\arctan\tfrac{x}{\sqrt{3}} + C}$$

Check. (a) Differentiating gives $x^2 - 2x^{-1/2} + \frac{5}{x}$, the integrand again. The $\ln|x|$ matters, because the integrand is defined for $x < 0$ too, and $\frac{d}{dx}\ln(-x) = \frac{1}{x}$ there. (b) Differentiating gives

$$3\sin\frac{x}{2} + 2\sqrt{3} \cdot \frac{1/\sqrt{3}}{1 + x^2/3}$$
$$= 3\sin\frac{x}{2} + \frac{6}{3+x^2}.$$

Pitfalls

- Reading a off a^2. Taking $a = 3$ in $3 + x^2$ gives $2\arctan\frac{x}{3}$, whose derivative is $\frac{6}{9+x^2}$, not $\frac{6}{3+x^2}$.
- Multiplying by a instead of dividing. $\int 3\sin\frac{x}{2}\,dx$ is not $-\frac{3}{2}\cos\frac{x}{2}$, whose derivative is $\frac{3}{4}\sin\frac{x}{2}$: with $a = \frac{1}{2}$, dividing by a doubles. Dropping the minus sign of the sin row is the other common slip: $6\cos\frac{x}{2}$ differentiates to $-3\sin\frac{x}{2}$.
- Writing $5\ln x$. It is undefined for $x < 0$, where $\frac{5}{x}$ is perfectly well defined. And the power rule cannot do $\frac{1}{x}$ at all: it would divide by $n + 1 = 0$.

In the notes. Unit 9: The antiderivative and the indefinite integral, for the power rule and $\ln|x|$, and Unit 9: A table of standard integrals.

Practise it. the Review set on this method

9.3 A definite integral by FTC 2; signed against total area

Problem. Evaluate $\int_a^b f(x)\,dx$ exactly; find the net signed area and the total area between a graph and the x-axis.

Method

1. Find an antiderivative F of f (Method 9.2). No $+C$ is needed.
2. Check that f is continuous on the whole of $[a,b]$ (a denominator that is 0 inside it rules this out). If it is not, FTC 2 does not apply: stop. Otherwise write $\big[F(x)\big]_a^b$, then $F(b)-F(a)$ with both values visible, and simplify.
3. For the total area, find where f changes sign in $[a,b]$.
4. Cut $[a,b]$ at the crossings and apply step 2 to each piece. The net signed area is the sum of the pieces; the total area is the sum of their magnitudes. State both.

Variants. A *net change*: the integral of a rate over $[a,b]$ is the change in the quantity, $\int_a^b Q'(t)\,dt = Q(b)-Q(a)$. Use step 2, or steps 3–4 applied to v for a total such as the distance travelled, $\int_a^b |v|\,dt$, and give the answer its unit.

Example. Find the net signed area and the total area between the graph of $f(x)=\dfrac{x-4}{\sqrt{x}}$ and the x-axis on $[1,9]$.

Split the quotient: $f(x)=\dfrac{x}{\sqrt{x}}-\dfrac{4}{\sqrt{x}}=x^{1/2}-4x^{-1/2}$. Term by term, $F(x)=\frac{2}{3}x^{3/2}-8x^{1/2}$, so

$$F(1)=\tfrac{2}{3}-8=-\tfrac{22}{3},\quad F(4)=\tfrac{16}{3}-16=-\tfrac{32}{3},\quad F(9)=18-24=-6.$$

The net signed area is

$$\big[F(x)\big]_1^9=-6-\left(-\frac{22}{3}\right)=\boxed{\tfrac{4}{3}}.$$

f changes sign only at $x=4$: it is negative on $[1,4)$ and positive on $(4,9]$. The two pieces are

$$\int_1^4 f\,dx=-\tfrac{32}{3}+\tfrac{22}{3}=-\tfrac{10}{3},\qquad \int_4^9 f\,dx=-6+\tfrac{32}{3}=\tfrac{14}{3},$$

and the total area is $\left|-\frac{10}{3}\right|+\frac{14}{3}=\boxed{8}$.

Check. Differentiate: $F'(x)=\frac{2}{3}\cdot\frac{3}{2}x^{1/2}-8\cdot\frac{1}{2}x^{-1/2}=x^{1/2}-4x^{-1/2}=f(x)$. The pieces add up to the net value, $-\frac{10}{3}+\frac{14}{3}=\frac{4}{3}$, and the first is negative, as the sign of f on $[1,4)$ says it must be.

Pitfalls

- Giving the net value $\frac{4}{3}$ as the area. The integral counts the part below the axis as negative; the area is 8.
- Taking $F(a)$ to be 0. It is not, unless the lower limit makes it so: here $F(1)=-\frac{22}{3}$, and leaving it out gives -6.
- Using FTC 2 where f is not continuous on $[a,b]$. For $\frac{1}{x}$ on $[-1,1]$ the bracket $\big[\ln|x|\big]_{-1}^{1}=0$ is a number, but the theorem does not apply, so it is not the integral.

In the notes. Unit 9: The antiderivative and the indefinite integral, Unit 9: The Fundamental Theorem of Calculus, and Unit 9: Properties of the definite integral, with its signed-area example.

Practise it. the Review set on this method

9.4 Differentiating an integral (FTC 1)

Problem. Differentiate, with respect to x, an integral $\int f(t)\,\mathrm{d}t$ whose limits depend on x but whose integrand does not, without evaluating it.

Method

1. Find which limit depends on x. If only the upper one does, go to step 3.
2. If the lower limit varies, reverse the limits, which flips the sign: $\int_{g(x)}^{b} f = -\int_{b}^{g(x)} f$. If both vary, split at a constant c: $\int_{g(x)}^{h(x)} f = \int_{c}^{h(x)} f - \int_{c}^{g(x)} f$.
3. In each piece $\int_{c}^{g(x)} f(t)\,\mathrm{d}t$, substitute the upper limit into the integrand: $f\big(g(x)\big)$.
4. Multiply by the derivative of that limit, $g'(x)$ (the chain rule). For the limit x itself, the factor is 1.
5. Combine the pieces with their signs. No antiderivative is needed, so this works even when there is no elementary one.

Why it works. With $F(u) = \int_{c}^{u} f(t)\,\mathrm{d}t$, FTC 1 says $F' = f$. The integral up to $g(x)$ is the composite $F\big(g(x)\big)$, and the chain rule gives $F'\big(g(x)\big)\,g'(x) = f\big(g(x)\big)\,g'(x)$.

Example. Find (a) $\dfrac{\mathrm{d}}{\mathrm{d}x}\displaystyle\int_{x}^{1} \frac{\sin t}{t}\,\mathrm{d}t$ for $x > 0$, and (b) $\dfrac{\mathrm{d}}{\mathrm{d}x}\displaystyle\int_{x}^{2x} \frac{\mathrm{d}t}{1+t^2}$.

(a) The variable is the lower limit. Reverse the limits: $\int_{x}^{1} \frac{\sin t}{t}\,\mathrm{d}t = -\int_{1}^{x} \frac{\sin t}{t}\,\mathrm{d}t$. Substitute x into the integrand; the factor is 1. So

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_{x}^{1} \frac{\sin t}{t}\,\mathrm{d}t = \boxed{-\frac{\sin x}{x}}.$$

(b) Both limits vary. Split at 0: $\int_{x}^{2x} = \int_{0}^{2x} - \int_{0}^{x}$. In the first piece substitute $2x$ and multiply by $(2x)' = 2$; the second gives $\frac{1}{1+x^2}$. So

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_{x}^{2x} \frac{\mathrm{d}t}{1+t^2} = \boxed{\frac{2}{1+4x^2} - \frac{1}{1+x^2}}.$$

Check. In (b) the integral can be evaluated first: $\big[\arctan t\big]_{x}^{2x} = \arctan 2x - \arctan x$, whose derivative is $\frac{2}{1+4x^2} - \frac{1}{1+x^2}$. In (a) it cannot, since $\frac{\sin t}{t}$ has no elementary antiderivative, but the sign can be checked: for $0 < x < \pi$ the integrand is positive, and moving the lower limit up takes a strip of height $\frac{\sin x}{x}$ away, so the derivative is negative.

Pitfalls

- Forgetting the chain-rule factor: the first piece of (b) is $\frac{2}{1+4x^2}$, not $\frac{1}{1+4x^2}$.
- Forgetting the sign when the variable is the lower limit: (a) is $-\frac{\sin x}{x}$, not $\frac{\sin x}{x}$.
- Letting the constant limit contribute: $\frac{\mathrm{d}}{\mathrm{d}x}\int_{1}^{x} f(t)\,\mathrm{d}t$ is $f(x)$, not $f(x) - f(1)$.

In the notes. Unit 9: The Fundamental Theorem of Calculus, from Part 1 to the example with a variable limit that is a function.

Practise it. the Review set on this method

9.5 Substitution, indefinite and definite

Problem. Integrate a function that is not in the table but contains an inside function together with its derivative; evaluate such a definite integral.

Method

1. Choose $u = g(x)$, the inside function whose derivative $g'(x)$ is also present, up to a constant factor.
2. Write $\mathrm{d}u = g'(x)\,\mathrm{d}x$, and solve it for the piece the integrand contains (for instance $x\,\mathrm{d}x = \frac{1}{2}\,\mathrm{d}u$).
3. For a definite integral, convert the limits: $x = a \Rightarrow u = g(a)$, $x = b \Rightarrow u = g(b)$.
4. Rewrite the whole integral in u. No x may be left. A constant may be moved across the integral sign; a function may not.
5. Integrate in u, from the table.
6. Indefinite: put $u = g(x)$ back and add $+C$. Definite: evaluate with the u-limits, and never return to x.
7. Check an indefinite answer by differentiating it.

Variants. *The f'/f pattern*: when the numerator is a constant c times the derivative of the denominator, $u =$ denominator gives $c\int \frac{\mathrm{d}u}{u}$, so $\int \frac{c\,g'(x)}{g(x)}\,\mathrm{d}x = c\ln|g(x)| + C$. Drop the bars only when $g > 0$ is known. *A stray x*: if $u = x + k$ leaves a factor x behind, write $x = u - k$ and expand into powers of u. *Completing the square*: a quadratic with no real roots in a denominator becomes $A\big[(x+h)^2 + a^2\big]$, with A its x^2 coefficient; take the $\frac{1}{A}$ out, put $u = x + h$ and use the table entry $\int \frac{\mathrm{d}u}{a^2+u^2} = \frac{1}{a}\arctan\frac{u}{a}$. *Route 1* for a definite integral: integrate, return to x, then apply the x-limits (Method 9.3). Both routes give the same number.

Example. Evaluate (a) $\displaystyle\int_0^{\pi/3} \frac{\sin x}{\cos^3 x}\,\mathrm{d}x$, and find (b) $\displaystyle\int \frac{\mathrm{d}x}{2x^2 + 12x + 36}$.

(a) The inside function is $u = \cos x$, and $\mathrm{d}u = -\sin x\,\mathrm{d}x$, so $\sin x\,\mathrm{d}x = -\mathrm{d}u$. The limits:

$$x = 0 \;\Rightarrow\; u = 1, \qquad x = \tfrac{\pi}{3} \;\Rightarrow\; u = \tfrac{1}{2}.$$

Hence

$$\begin{aligned}\int_0^{\pi/3} \frac{\sin x}{\cos^3 x}\,\mathrm{d}x &= -\int_1^{1/2} u^{-3}\,\mathrm{d}u \\ &= \int_{1/2}^{1} u^{-3}\,\mathrm{d}u = \Big[-\frac{1}{2u^2}\Big]_{1/2}^{1} \\ &= -\frac{1}{2} - (-2) = \boxed{\tfrac{3}{2}}.\end{aligned}$$

(b) The denominator has no real roots. Take out the 2 and complete the square: $2x^2+12x+36 = 2\big[(x+3)^2+9\big]$. Put $u = x + 3$, so $\mathrm{d}u = \mathrm{d}x$, and use the table with $a = 3$:

$$\tfrac{1}{2}\int \frac{\mathrm{d}u}{9+u^2} = \tfrac{1}{6}\arctan\frac{u}{3} + C = \boxed{\tfrac{1}{6}\arctan\frac{x+3}{3} + C}.$$

Check. (a) by Route 1: $-\int u^{-3}\,du = \frac{1}{2u^2}$, so an antiderivative is $\frac{1}{2\cos^2 x}$. Its derivative is $\frac{\sin x}{\cos^3 x}$, the integrand; with the minus sign lost it would be minus the integrand. And

$$\left[\frac{1}{2\cos^2 x}\right]_0^{\pi/3} = 2 - \tfrac{1}{2} = \tfrac{3}{2}.$$

(b) Differentiate:

$$\tfrac{1}{6}\cdot\frac{1/3}{1+(x+3)^2/9} = \frac{1}{18+2(x+3)^2} = \frac{1}{2x^2+12x+36}.$$

Pitfalls

- Keeping the x-limits after changing the variable, as in $\int_0^{\pi/3} u^{-3}\,du$. The limits belong to the variable of integration.
- Losing the minus sign of $du = -\sin x\,dx$. It gives $-\frac{3}{2}$, a negative value for an integrand that is positive on $(0, \frac{\pi}{3})$.
- Losing a factor. Dropping the $\frac{1}{a}$ of the table entry gives $\frac{1}{2}\arctan\frac{x+3}{3}$; dropping the $\frac{1}{2}$ taken out gives $\frac{1}{3}\arctan\frac{x+3}{3}$. They differentiate to $\dfrac{3}{2x^2+12x+36}$ and $\dfrac{2}{2x^2+12x+36}$.

In the notes. Unit 9: Integration by substitution, Unit 9: Substitution in a definite integral, and Unit 9: A table of standard integrals.

Practise it. the Review set on this method

9.6 No elementary antiderivative: decide, then bracket

Problem. Decide whether FTC 2 can evaluate an integral with an elementary antiderivative. When it cannot, say why, say what antiderivative there is, and estimate the definite integral from Riemann sums.

Method

1. Look for an inside function whose derivative is also present, up to a constant. If there is one, the integral is elementary: substitute (Method 9.5) and evaluate by FTC 2.
2. If the integrand is one the notes name, e^{-x^2}, $\frac{\sin x}{x}$, $\sin(x^2)$, $\frac{e^x}{x}$ or $\frac{1}{\ln x}$, it has no elementary antiderivative (Liouville). Quote this. It still has an antiderivative, its area function $F(x) = \int_a^x f(t)\,dt$ (FTC 1). What it lacks is a formula, so FTC 2 has nothing to substitute into.
3. Estimate the definite integral by the left and right sums with n strips (Method 9.1, steps 1–2).
4. Decide the direction of each sum from monotonicity (Method 9.1, step 3), and state the bracket, rounding outwards. More strips tighten it.

Example. Of $P = \displaystyle\int_2^4 \frac{dx}{\ln x}$ and $Q = \displaystyle\int_2^4 \frac{dx}{x\ln x}$, one can be evaluated by FTC 2. Evaluate it. For the other, say why FTC 2 cannot, and bracket it with $n = 4$ strips.

Q: the inside function is $u = \ln x$, and its derivative $\frac{1}{x}$ is present. So $du = \frac{dx}{x}$, and the

limits are

$$x = 2 \;\Rightarrow\; u = \ln 2, \qquad x = 4 \;\Rightarrow\; u = \ln 4 = 2\ln 2.$$

Hence

$$\begin{aligned} Q &= \int_{\ln 2}^{2\ln 2} \frac{\mathrm{d}u}{u} \\ &= \ln(2\ln 2) - \ln(\ln 2) \\ &= \boxed{\ln 2} \approx 0.693. \end{aligned}$$

P: $\frac{1}{\ln x}$ is on the notes' list, and no factor $\frac{1}{x}$ is present. (Forcing $u = \ln x$ gives $\mathrm{d}x = \mathrm{e}^u\,\mathrm{d}u$ and turns P into $\int \frac{\mathrm{e}^u}{u}\,\mathrm{d}u$, another entry on the list.) So there is no elementary antiderivative, only the area function $F(x) = \int_2^x \frac{\mathrm{d}t}{\ln t}$. With $\Delta x = 0.5$, the heights at $x = 2,\ 2.5,\ 3,\ 3.5,\ 4$ are $1.4427,\ 1.0914,\ 0.9102,\ 0.7982,\ 0.7213$. The first four add to 4.2425 and the last four to 3.5211, so

$$\text{left} = 0.5 \cdot 4.2425 \approx 2.1213,$$

$$\text{right} = 0.5 \cdot 3.5211 \approx 1.7606.$$

$\ln x$ increases and is positive on $[2, 4]$, so $\frac{1}{\ln x}$ decreases: the left sum is the over-estimate. Rounding outwards,

$$\boxed{1.760 < P < 2.122}$$

Check. Q: $\frac{\mathrm{d}}{\mathrm{d}x}\ln(\ln x) = \frac{1}{\ln x}\cdot\frac{1}{x}$, the integrand, and $\ln 2 > 0$, as a positive integrand requires. P: the two sums share the three middle heights, so they differ by $0.5\,(f(2) - f(4)) = 0.5\,(1.4427 - 0.7213) \approx 0.3607$, and they do $(2.1213 - 1.7606)$. A computer gives $P \approx 1.9224$, inside the bracket.

Pitfalls

- Reading "no elementary antiderivative" as "no antiderivative". P has one, $F(x) = \int_2^x \frac{\mathrm{d}t}{\ln t}$ with $F'(x) = \frac{1}{\ln x}$. What it lacks is a formula.
- Dividing by the derivative of the inside. $\int \mathrm{e}^{-x^2}\,\mathrm{d}x$ is not $\frac{\mathrm{e}^{-x^2}}{-2x}$: its derivative is $\mathrm{e}^{-x^2}\left(1 + \frac{1}{2x^2}\right)$. Substitution needs the derivative present as a factor; you cannot divide by it.
- Saying "no" because a substitution fails. Only the functions the notes list may be quoted as non-elementary. $x\,\mathrm{e}^x$ offers no substitution, yet it integrates, by integration by parts, in Unit 10.

In the notes. Unit 9: When there is no elementary antiderivative, Unit 9: The Fundamental Theorem of Calculus (Part 1), and Unit 9: The area under a curve, for the bracket.

Practise it. the Review set on this method

Unit 10 Integration II: techniques and applications

Seven methods: four techniques of integration, then three uses of the integral. On this unit choosing the technique is most of the work. Look for a substitution first (Method 9.5); when there is none, the shape of the integrand says what to try next.

Which method?

The integrand, or the question, has	Method
an inside function times its derivative; an f'/f	9.5, substitution
a polynomial times e^{ax}, $\sin ax$, $\cos ax$ or $\sec^2 ax$	10.1, by parts, u = the polynomial
a polynomial times $\ln x$ or $\arctan x$	10.1, by parts, $u = \ln x$ or $\arctan x$
a lone $\ln x$ or $\arctan x$	10.1, by parts with $v' = 1$
e^{ax} times $\sin bx$ or $\cos bx$	10.1, parts twice, then solve for the integral
a polynomial over a polynomial, with no repeated irreducible quadratic factor	10.2: divide if improper, then factor
1 over a quadratic with no real roots	10.2: complete the square, then arctan
an odd power of $\sin x$ or of $\cos x$	10.3: split off one factor
only even powers of $\sin x$ and $\cos x$	10.3: the half-angle identities
$\tan^m x \sec^n x$ with $n \geq 2$ even or m odd; a power of $\tan x$ alone	10.3: peel off $\sec^2 x$ if you can, else $\tan^2 x$
$\sqrt{a^2 - x^2}$	10.4, $x = a \sin\theta$
$\sqrt{a^2 + x^2}$, or a power of $a^2 + x^2$, such as $1/(a^2 + x^2)^2$	10.4, $x = a \tan\theta$
$\sqrt{x^2 - a^2}$	10.4, $x = a \sec\theta$
a limit of integration ∞ or $-\infty$	10.5, improper integral
a finite interval with a point where the integrand is unbounded	10.5, a limit at the blow-up; split at an interior one
an area, a volume, work, a mass or an average value, in words	10.6, slice and sum; an average: its Variants
a density: its constant, a probability, a mean	10.7, density

10.1 Integration by parts

Problem. Integrate a product of two different kinds of function, such as a polynomial times an exponential, or a lone logarithm or inverse trigonometric function, such as $\ln x$.

Method

1. Name the choice by LIATE: u is the factor that comes first in Logarithm, Inverse trig, Algebraic, Trigonometric, Exponential, and the rest is v'. For a lone function such as $\ln x$ or $\arctan x$, take $v' = 1$.
2. Write down u' and v.
3. Apply $\int uv'\,dx = uv - \int u'v\,dx$. For a definite integral the first term is $\left[uv\right]_a^b$.
4. Evaluate the new integral. If it needs parts again, repeat with the same kind of choice.

5. If the integral you started with comes back, call it I and solve for I algebraically. Do not keep going.
6. Add $+C$ to an indefinite result.

Example. Evaluate (a) $\displaystyle\int_0^{\pi/4} x\sec^2 x\,\mathrm{d}x$ and (b) $\displaystyle\int \sin(\ln x)\,\mathrm{d}x$ for $x>0$.

(a) LIATE: x (algebraic) before $\sec^2 x$ (trigonometric), so $u=x$, $v'=\sec^2 x$, giving $u'=1$ and $v=\tan x$. Call the integral J. Then

$$\begin{aligned} J &= \big[x\tan x\big]_0^{\pi/4} - \int_0^{\pi/4} \tan x\,\mathrm{d}x \\ &= \tfrac{\pi}{4} - \big[\ln|\sec x|\big]_0^{\pi/4} = \tfrac{\pi}{4} - \ln\sqrt{2}, \end{aligned}$$

that is, $\boxed{\tfrac{\pi}{4}-\tfrac{1}{2}\ln 2}\approx 0.439$.

(b) A lone function, so try $v'=1$, as for $\ln x$: take $u=\sin(\ln x)$, $v'=1$: then $u'=\cos(\ln x)/x$ and $v=x$, and the x cancels:

$$I=\int \sin(\ln x)\,\mathrm{d}x = x\sin(\ln x) - \int \cos(\ln x)\,\mathrm{d}x.$$

Parts again, the same way ($u=\cos(\ln x)$, $v'=1$, $u'=-\sin(\ln x)/x$):

$$\int \cos(\ln x)\,\mathrm{d}x = x\cos(\ln x) + I.$$

The original integral has come back. Substitute and solve for I:

$$I = x\sin(\ln x) - x\cos(\ln x) - I,$$

so $2I = x\big(\sin(\ln x)-\cos(\ln x)\big)$ and

$$I=\boxed{\tfrac{x}{2}\big(\sin(\ln x)-\cos(\ln x)\big)+C}$$

Check. Differentiate. In (a), $\frac{\mathrm{d}}{\mathrm{d}x}\big(x\tan x-\ln|\sec x|\big)=\tan x+x\sec^2 x-\tan x=x\sec^2 x$. In (b), the product rule gives $\frac{1}{2}\big(\sin(\ln x)-\cos(\ln x)\big)+\frac{1}{2}\big(\cos(\ln x)+\sin(\ln x)\big)=\sin(\ln x)$.

Pitfalls

- Choosing the other way round. In (a), $u=\sec^2 x$ and $v'=x$ give $\int \frac{1}{2}x^2\cdot 2\sec^2 x\tan x\,\mathrm{d}x$, worse than the integral you started with.
- Applying parts a third time in (b). Once I reappears, the next round only undoes the last one; solve for I instead.
- Dropping the minus sign in front of $\int u'v\,\mathrm{d}x$, or in $u'=-\sin(\ln x)/x$. In (b) that turns $-I$ into $+I$, and I cancels from the equation.

In the notes. Unit 10: Integration by parts, with LIATE and its three examples.

Practise it. the Review set on this method

10.2 Partial fractions, branched by the denominator

Problem. Integrate a rational function $\dfrac{P(x)}{Q(x)}$, a polynomial over a polynomial. A power of an irreducible quadratic, such as $1/(x^2+1)^2$, is not split here: use Method 10.4.

Method

1. Is it proper? If the degree of P is at least that of Q, divide first: a polynomial plus a proper remainder.
2. Factor Q into linear factors and irreducible quadratics (negative discriminant). If Q is a single irreducible quadratic and P is a constant, there is nothing to split: complete the square, $(x+p)^2+q$, and put $u=x+p$.
3. Write the template: $\dfrac{A}{ax+b}$ for each linear factor; one term for each power of a repeated linear factor, up to the highest; and $\dfrac{Ax+B}{ax^2+bx+c}$ for an irreducible quadratic.
4. Clear the denominators. Cover-up: put x equal to the root of each linear factor, which wipes out every term but one. Match a coefficient (or substitute another x) for each constant cover-up cannot reach.
5. Integrate term by term: $\dfrac{A}{ax+b}$ gives $\frac{A}{a}\ln|ax+b|$, and $\dfrac{1}{(ax+b)^2}$ gives $-\dfrac{1}{a(ax+b)}$. Over an irreducible quadratic, split the numerator into the f'/f part (a ln) and the rest (an arctan, after completing the square).

Example. Evaluate (a) $\displaystyle\int \frac{x^2+2}{x^2+x-2}\,\mathrm{d}x$ and (b) $\displaystyle\int \frac{2x^2+x+2}{(x-1)(x^2+4)}\,\mathrm{d}x$.

(a) Both degrees are 2, so divide first:

$$\frac{x^2+2}{x^2+x-2} = 1 + \frac{4-x}{x^2+x-2}.$$

The denominator factors as $(x-1)(x+2)$, two distinct linear factors, so

$$\frac{4-x}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2},$$

and $4-x = A(x+2)+B(x-1)$. Cover-up: $x=1$ gives $3=3A$, so $A=1$; $x=-2$ gives $6=-3B$, so $B=-2$. Integrating term by term,

$$\boxed{x + \ln|x-1| - 2\ln|x+2| + C}$$

(b) Proper, with a linear factor and the quadratic x^2+4, which is irreducible (discriminant -16). The template is

$$\frac{2x^2+x+2}{(x-1)(x^2+4)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+4},$$

so $2x^2+x+2 = A(x^2+4)+(Bx+C)(x-1)$. Cover-up at $x=1$: $5=5A$, so $A=1$. Cover-up cannot reach B and C. Match coefficients instead: the x^2 terms give $2=A+B$, so $B=1$; the

constants give $2 = 4A - C$, so $C = 2$. Split the quadratic term into its f'/f part and the rest:

$$\frac{x+2}{x^2+4} = \frac{1}{2}\cdot\frac{2x}{x^2+4} + \frac{2}{x^2+4}.$$

With $\int \frac{dx}{a^2+x^2} = \frac{1}{a}\arctan\frac{x}{a}$ and $a = 2$,

$$\boxed{\ln|x-1| + \tfrac{1}{2}\ln(x^2+4) + \arctan\tfrac{x}{2} + C}$$

Check. Recombine at a convenient x. At $x = 0$, the integrand of (a) is $\frac{2}{-2} = -1$ and the split form gives $1 + \frac{1}{-1} - \frac{2}{2} = -1$; the integrand of (b) is $\frac{2}{(-1)\cdot 4} = -\frac{1}{2}$ and the split form gives $\frac{1}{-1} + \frac{2}{4} = -\frac{1}{2}$. But $x = 0$ wipes out the Bx term, so test (b) at $x = 2$ too: $\frac{12}{1\cdot 8} = \frac{3}{2}$ and $\frac{1}{1} + \frac{4}{8} = \frac{3}{2}$.

Pitfalls

- Splitting before dividing. In (a), the template $\frac{A}{x-1} + \frac{B}{x+2}$ can never produce the 1 that division gives, and the equations for A and B are inconsistent.
- Putting a constant over an irreducible quadratic. In (b), the numerator over x^2+4 must be $Bx + C$, with two constants.
- Reaching for partial fractions when the numerator is the derivative of the denominator: $\int \frac{2x+1}{x^2+x-2}\,dx$ is $\ln|x^2+x-2| + C$ at once, by substitution.

In the notes. Unit 10: Partial fractions, with the recipe and its four examples.

Practise it. the Review set on this method

10.3 Trigonometric integrals: odd or even powers, tan and sec

Problem. Integrate a product of powers of $\sin x$ and $\cos x$, or of $\tan x$ and $\sec x$.

Method

1. Classify. Powers of sin and cos: if one power is odd, split off one factor of it; if both are even, use the half-angle identities. Powers $\tan^m x \sec^n x$: if $n \geq 2$ is even, peel off $\sec^2 x$; otherwise, if m is odd or there is no $\sec x$, peel off $\tan^2 x = \sec^2 x - 1$.
2. Rewrite the rest with an identity: $\sin^2 x + \cos^2 x = 1$ for an odd power; $\cos^2 x = \frac{1}{2}(1+\cos 2x)$ and $\sin^2 x = \frac{1}{2}(1-\cos 2x)$ for even powers; $1 + \tan^2 x = \sec^2 x$ for tangents and secants.
3. Substitute: $u = \cos x$ if you split off $\sin x$ ($du = -\sin x\,dx$), $u = \sin x$ if you split off $\cos x$, $u = \tan x$ for each piece $\tan^k x \sec^2 x$, $u = \sec x$ for a piece $\sec^k x \sec x \tan x$. For a definite integral, map the limits, and keep the sign.
4. A leftover $\int \tan x\,dx$ is $\ln|\sec x| + C$.
5. Write the result in terms of x, and add $+C$ to an indefinite one.

Example. Evaluate (a) $\displaystyle\int_0^{\pi/2} \sin^5 x \cos^3 x\,dx$ and (b) $\displaystyle\int \tan^3 x \sec^4 x\,dx$.

(a) Both powers are odd, so either can be split. Split off one $\cos x$ and write the rest in sines: $\sin^5 x \cos^3 x = \sin^5 x\,(1 - \sin^2 x)\cos x$. Put $u = \sin x$, $du = \cos x\,dx$, with $x = 0 \mapsto u = 0$

and $x = \frac{\pi}{2} \mapsto u = 1$:

$$\int_0^{\pi/2} \sin^5 x \cos^3 x \, dx = \int_0^1 u^5(1-u^2)\, du$$
$$= \tfrac{1}{6} - \tfrac{1}{8} = \boxed{\tfrac{1}{24}}.$$

(b) The power of sec is even, so peel off $\sec^2 x$ (m is odd too, but n comes first) and convert the rest, $\tan^3 x \sec^4 x = \tan^3 x\,(1+\tan^2 x)\sec^2 x$. With $u = \tan x$, $du = \sec^2 x \, dx$,

$$\int \tan^3 x \sec^4 x \, dx = \int u^3(1+u^2)\, du = \tfrac{1}{4}u^4 + \tfrac{1}{6}u^6 + C,$$

that is, $\boxed{\tfrac{1}{4}\tan^4 x + \tfrac{1}{6}\tan^6 x + C}$.

Check. In (a), split off $\sin x$ instead: $u = \cos x$ maps $0 \mapsto 1$ and $\frac{\pi}{2} \mapsto 0$, and the minus sign from du turns the limits round, giving $\int_0^1 (1-u^2)^2 u^3 \, du = \frac{1}{4} - \frac{1}{3} + \frac{1}{8} = \frac{1}{24}$ again. In (b), differentiate: $\tan^3 x \sec^2 x + \tan^5 x \sec^2 x = \tan^3 x \sec^2 x\,(1+\tan^2 x) = \tan^3 x \sec^4 x$.

Pitfalls

- Using the half-angle identity on an odd power. It works, but slowly; split off one factor instead.
- Losing the minus sign of $du = -\sin x \, dx$. With $u = \cos x$ in (a) it gives $-\frac{1}{24}$: a negative answer for an integrand that is positive on $(0, \frac{\pi}{2})$.
- Converting every power of sec. In (b), $\sec^4 x = (1+\tan^2 x)^2$ leaves no $\sec^2 x$ to make du.

In the notes. Unit 10: Trigonometric integrals.
Practise it. the Review set on this method

10.4 Trigonometric substitution

Problem. Integrate an expression containing $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$ or $\sqrt{x^2 - a^2}$ (or a power of $a^2 + x^2$) that no plain substitution clears.

Method

1. Read off the form and $a > 0$, and substitute: $\sqrt{a^2 - x^2}$: $x = a\sin\theta$, $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$; $\sqrt{a^2 + x^2}$: $x = a\tan\theta$, $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$; $\sqrt{x^2 - a^2}$ with $x \geq a$: $x = a\sec\theta$, $\theta \in [0, \frac{\pi}{2})$.
2. Write dx, and simplify the root by a Pythagorean identity. Use the range of θ to drop the modulus: $\sqrt{a^2 - a^2\sin^2\theta} = a\,|\cos\theta| = a\cos\theta$.
3. Integrate in θ, often a trigonometric integral (Method 10.3). For a definite integral, change the limits to θ and finish there.
4. For an indefinite integral, convert back to x: through the substitution and a right triangle, or algebraically (for instance $\cos\theta = \sqrt{1 - \sin^2\theta}$ on the range).
5. Add $+C$ to an indefinite result.

Example. Evaluate (a) $\displaystyle\int \frac{dx}{(x^2+1)^2}$ and (b) $\displaystyle\int_0^1 \frac{x^2}{\sqrt{4-x^2}}\,dx$.

(a) A power of a^2+x^2 with $a=1$: put $x=\tan\theta$, $\theta\in(-\tfrac{\pi}{2},\tfrac{\pi}{2})$, so $dx=\sec^2\theta\,d\theta$ and $(x^2+1)^2=\sec^4\theta$. Then, by the half-angle identity,

$$\int \frac{\sec^2\theta}{\sec^4\theta}\,d\theta=\int \cos^2\theta\,d\theta=\tfrac{1}{2}\theta+\tfrac{1}{4}\sin 2\theta+C.$$

Convert back: $\theta=\arctan x$, and $\sin 2\theta=2\sin\theta\cos\theta=\dfrac{2\tan\theta}{\sec^2\theta}=\dfrac{2x}{1+x^2}$. So the integral is

$$\boxed{\tfrac{1}{2}\arctan x+\frac{x}{2(1+x^2)}+C}$$

(b) The form $\sqrt{a^2-x^2}$ with $a=2$: put $x=2\sin\theta$, $\theta\in[-\tfrac{\pi}{2},\tfrac{\pi}{2}]$, so $dx=2\cos\theta\,d\theta$ and $\sqrt{4-x^2}=2\cos\theta$. The limits: $x=0\mapsto\theta=0$, and $x=1\mapsto\sin\theta=\tfrac{1}{2}$, so $\theta=\tfrac{\pi}{6}$. Then

$$\begin{aligned}\int_0^1 \frac{x^2\,dx}{\sqrt{4-x^2}}&=\int_0^{\pi/6}\frac{4\sin^2\theta\cdot 2\cos\theta}{2\cos\theta}\,d\theta\\&=\int_0^{\pi/6}4\sin^2\theta\,d\theta\\&=\int_0^{\pi/6}2(1-\cos 2\theta)\,d\theta\\&=\big[2\theta-\sin 2\theta\big]_0^{\pi/6},\end{aligned}$$

which is $\boxed{\tfrac{\pi}{3}-\tfrac{\sqrt{3}}{2}}\approx 0.181$.

Check. In (a), differentiate, using the quotient rule on the second term:

$$\frac{1}{2(1+x^2)}+\frac{1-x^2}{2(1+x^2)^2}=\frac{1}{(1+x^2)^2}.$$

In (b), the integrand increases from 0 to $\tfrac{1}{\sqrt{3}}\approx 0.577$ on $[0,1]$, so the value must lie between 0 and 0.577; it does.

Pitfalls

- Choosing by habit: $\sqrt{4-x^2}$ takes sin, $\sqrt{x^2-4}$ takes sec. With the wrong one no identity clears the root: $x=2\sin\theta$ makes $x^2-4=-4\cos^2\theta$, which is negative.
- Forgetting to change the limits in (b), or changing them and then converting back to x as well.
- Using a trigonometric substitution where a plain one works: $\int x\sqrt{4-x^2}\,dx$ needs only $u=4-x^2$.

In the notes. Unit 10: Trigonometric substitution, with its sine and secant examples and the right triangle.

Practise it. the Review set on this method

10.5 Improper integrals

Problem. Evaluate an integral over an unbounded interval, or of a function that is unbounded on a finite interval, or show that it diverges.

Method

1. Replace ∞ by t (and $-\infty$ by s). Split a doubly infinite integral at any convenient point c into two pieces.
2. Integrate on the finite interval, $[a, t]$, as an ordinary definite integral.
3. Take the limit $t \to \infty$. Justify each term: a term such as $t\,\mathrm{e}^{-t}$ tends to 0 by l'Hôpital's rule (Unit 8).
4. State the verdict: the integral converges to the limit, or it diverges. A split integral converges only if each piece does.

Example. Evaluate, or show divergent: (a) $\displaystyle\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{x^2-6x+13}$ and (b) $\displaystyle\int_{2}^{\infty} \frac{x}{x^2+1}\,\mathrm{d}x$.

(a) Complete the square: $x^2-6x+13=(x-3)^2+4$, so an antiderivative is $\frac{1}{2}\arctan\frac{x-3}{2}$. Split at $c=3$:

$$\int_3^t \frac{\mathrm{d}x}{x^2-6x+13} = \tfrac{1}{2}\arctan\tfrac{t-3}{2} \to \tfrac{\pi}{4},$$
$$\int_s^3 \frac{\mathrm{d}x}{x^2-6x+13} = -\tfrac{1}{2}\arctan\tfrac{s-3}{2} \to \tfrac{\pi}{4},$$

as $t \to \infty$ and $s \to -\infty$. Both pieces converge, so the integral converges to $\frac{\pi}{4}+\frac{\pi}{4}=\boxed{\frac{\pi}{2}}$.

(b) The numerator is half the derivative of the denominator:

$$\int_2^t \frac{x}{x^2+1}\,\mathrm{d}x = \Big[\tfrac{1}{2}\ln(x^2+1)\Big]_2^t = \tfrac{1}{2}\ln(t^2+1)-\tfrac{1}{2}\ln 5.$$

This grows without bound as $t \to \infty$, so the integral $\boxed{\text{diverges}}$.

Variants. An integrand that is unbounded at a point of a finite interval makes the integral improper too. At an endpoint a, integrate from s and let $s \to a^+$; at an interior point c, split at c and require both halves to converge, since FTC 2 applied straight across the blow-up can return a finite but wrong answer. For instance,

$$\int_{-1}^{8} \frac{\mathrm{d}x}{\sqrt[3]{x^2}} = \lim_{t\to 0^-}\Big[3\sqrt[3]{x}\Big]_{-1}^{t} + \lim_{s\to 0^+}\Big[3\sqrt[3]{x}\Big]_{s}^{8} = 3+6=9.$$

Check. In (a), differentiate the antiderivative: the chain rule gives $\frac{1}{2}\cdot\frac{1}{2}\cdot 1/\big(1+\frac{1}{4}(x-3)^2\big) = 1/\big((x-3)^2+4\big)$, the integrand; a missing $\frac{1}{2}$ would double it. In (b), $\frac{x}{x^2+1} \geq \frac{1}{2x}$ for $x \geq 1$ (it rearranges to $x^2 \geq 1$), and the tail of $\frac{1}{x}$ has infinite area, so divergence is no surprise.

Pitfalls

- Putting ∞ into the formula as if it were a number. $\left[\cdot\right]_2^{\infty}$ is only shorthand for the limit as $t \to \infty$, taken after the integral is evaluated, and it has to be justified.
- Deciding from the integrand alone. The integrand of (b) tends to 0, yet the integral diverges.
- Evaluating a doubly infinite integral as one symmetric limit, $\lim_{t\to\infty} \int_{-t}^{t}$. That can be finite when the two halves diverge; take the pieces separately.

In the notes. Unit 10: Improper integrals, probability and expectation, its definition and first example.

Practise it. the Review set on this method

10.6 Setting up an integral by slice and sum

Problem. Turn a description into an integral: the area between two curves, the volume of a solid of revolution or of known cross-section, the work done lifting something, or the mass of a rod.

Method

1. Say what x measures, from where, and the range it covers. For an area, find where the curves meet; sketch if asked.
2. Write the contribution of one slice of width $\mathrm{d}x$ as (something) $\times\ \mathrm{d}x$. A strip of height (top − bottom), with the top curve checked by a test point. A disc of area $\pi f(x)^2$. A slab of area $A(x)$, by similar triangles if need be. A mass $\mathrm{d}m = \rho(x)\ \mathrm{d}x$. Work $\mathrm{d}W$ = force × distance.
3. Integrate over the range. For an area, split at every crossing, so that each piece has the right curve on top.
4. Evaluate, and give the units.
5. Check against a known formula or a simple bound.

Variants. The *average value* of f on $[a, b]$ is $\bar{f} = \frac{1}{b-a}\int_a^b f(x)\ \mathrm{d}x$, in the units of f: the height of the rectangle with the same area.

Example. (a) Find the area enclosed between $y = x^3$ and $y = 4x$. (b) The region under $y = \dfrac{1}{\sqrt{1+x^2}}$, $0 \le x \le 1$, is revolved about the x-axis. Find the volume. (c) A cylindrical tank of radius 1 m and height 2 m is full of water, of density $1000\,\mathrm{kg\,m^{-3}}$. Find the work done pumping all of it out over the top. Take $g = 9.8\,\mathrm{m\,s^{-2}}$.

(a) The curves meet where $x^3 = 4x$, that is $x(x-2)(x+2) = 0$: at $x = -2, 0, 2$. A test point in each interval: at $x = -1$, $x^3 = -1 > -4$, so x^3 is on top on $[-2, 0]$; at $x = 1$, $4x = 4 > 1$, so $4x$ is on top on $[0, 2]$. Split at 0:

$$\begin{aligned} A &= \int_{-2}^{0} (x^3 - 4x)\ \mathrm{d}x + \int_{0}^{2} (4x - x^3)\ \mathrm{d}x \\ &= \left[\tfrac{1}{4}x^4 - 2x^2\right]_{-2}^{0} + \left[2x^2 - \tfrac{1}{4}x^4\right]_{0}^{2} \\ &= 4 + 4, \end{aligned}$$

so $\boxed{A = 8}$.

(b) The disc at x has radius $f(x) = \dfrac{1}{\sqrt{1+x^2}}$ and area $\pi f(x)^2 = \dfrac{\pi}{1+x^2}$, so

$$V = \pi \int_0^1 \frac{\mathrm{d}x}{1+x^2} = \pi \left[\arctan x\right]_0^1 = \boxed{\tfrac{\pi^2}{4}} \approx 2.47.$$

(c) Let x (m) be the depth below the top, $0 \le x \le 2$. The slice at depth x has volume $\pi \cdot 1^2 \, \mathrm{d}x$, so mass $1000\pi \, \mathrm{d}x$ and weight $9800\pi \, \mathrm{d}x$ (N), and it is raised a distance x: $\mathrm{d}W = 9800\pi x \, \mathrm{d}x$. Then

$$W = \int_0^2 9800\pi x \, \mathrm{d}x = 9800\pi \left[\tfrac{1}{2}x^2\right]_0^2,$$

so $\boxed{W = 19600\pi \text{ J}} \approx 6.16 \times 10^4$ J (joules: 1 J is 1 N m).

Check. (a) On $[0, 2]$ the piece is the triangle under $y = 4x$, of area $\frac{1}{2} \cdot 2 \cdot 8 = 8$, less the area $\int_0^2 x^3 \, \mathrm{d}x = 4$ under $y = x^3$: 4. Both curves are odd, so a half-turn about the origin carries it onto the other piece: 4 each. (b) One disc at the midpoint, with $f(\frac{1}{2})^2 = \frac{4}{5}$ and width 1, estimates $V \approx \frac{4\pi}{5} \approx 2.51$, close to 2.47; forgetting to square gives $\pi \ln(1 + \sqrt{2}) \approx 2.77$, far off. (c) The water weighs $2000\pi \cdot 9.8 = 19600\pi$ N and its centre is 1 m below the top: weight times that depth is 19600π J.

Pitfalls

- Integrating across a crossing. In (a), $\int_{-2}^{2} (4x - x^3) \, \mathrm{d}x = 0$: the two pieces cancel, but the area is 8.
- Forgetting to square the radius: $\pi \int_0^1 \frac{\mathrm{d}x}{\sqrt{1+x^2}}$ is not the volume in (b).
- Measuring x from the bottom of the tank in (c) but keeping the distance x: a slice at height x above the bottom is raised $2 - x$.

In the notes. Unit 10: Area between two curves, Unit 10: Volumes of revolution, Unit 10: The average value of a function (the Variant), and Unit 10: Setting up integrals from a description.

Practise it. the Review set on this method For the average value, the Review set on this variant

10.7 Probability density: normalise, probability, expectation

Problem. A random quantity X has a density f, perhaps with an unknown constant. Fix the constant, then find a probability and the expectation.

Method

1. Check that f is a density: $f(x) \ge 0$, and $\int f = 1$ over its range. If f contains an unknown constant k, impose $\int f = 1$ to find it.
2. A probability is an area: $P(a \le X \le b) = \int_a^b f(x) \, \mathrm{d}x$. Over an infinite range the integral is improper (Method 10.5).
3. The expectation is $\mathbb{E}[X] = \int x \, f(x) \, \mathrm{d}x$ over the whole range. Over an infinite range it is improper too, and it must converge.
4. Check: the probabilities of an event and of its complement add to 1, and $\mathbb{E}[X]$ lies between the smallest and largest values X can take.

Example. X has density $f(x) = \dfrac{k}{(1+x)^3}$ on $[0,\infty)$, and 0 elsewhere. Find k, $P(X>1)$ and $\mathbb{E}[X]$.

Normalise. An antiderivative of $(1+x)^{-3}$ is $-\frac{1}{2}(1+x)^{-2}$, so

$$\int_0^t \frac{k\,dx}{(1+x)^3} = \frac{k}{2}\Big(1 - \frac{1}{(1+t)^2}\Big),$$

which tends to $\frac{k}{2}$ as $t\to\infty$. Setting $\frac{k}{2}=1$ gives $\boxed{k=2}$, and then $f\ge 0$.

The probability is an improper integral:

$$P(X>1) = \lim_{t\to\infty}\Big[-\frac{1}{(1+x)^2}\Big]_1^t = 0 + \frac{1}{4} = \boxed{\tfrac{1}{4}}.$$

For the mean, write $x = (1+x)-1$ in the numerator:

$$\frac{2x}{(1+x)^3} = \frac{2}{(1+x)^2} - \frac{2}{(1+x)^3}.$$

Then

$$\begin{aligned}\int_0^t \frac{2x\,dx}{(1+x)^3} &= \Big[-\frac{2}{1+x} + \frac{1}{(1+x)^2}\Big]_0^t \\ &= -\frac{2}{1+t} + \frac{1}{(1+t)^2} + 1,\end{aligned}$$

which tends to 1 as $t\to\infty$: $\boxed{\mathbb{E}[X]=1}$.

Check. $P(X\le 1) = \big[-\frac{1}{(1+x)^2}\big]_0^1 = 1-\frac{1}{4} = \frac{3}{4}$, and $\frac{3}{4}+\frac{1}{4}=1$. For the mean, $\mathbb{E}[X]\ge 0$ is a weak test, and 1 is also the value of $\int f$, the first Pitfall's wrong answer, so differentiate instead: $\frac{d}{dx}\big(-\frac{2}{1+x}+\frac{1}{(1+x)^2}\big) = \frac{2}{(1+x)^2} - \frac{2}{(1+x)^3}$, the split integrand.

Pitfalls

- Finding $\mathbb{E}[X]$ from $\int f$ instead of $\int xf$. The first is 1 for every density.
- Taking $P(X>1)$ over $[0,1]$. Draw the interval the event describes first.
- Using k before fixing it, or fixing it from $\int_0^1 f = 1$ when the range is $[0,\infty)$.

In the notes. Unit 10: Improper integrals, probability and expectation, from "Probability densities and expectation" on.

Practise it. the Review set on this method